

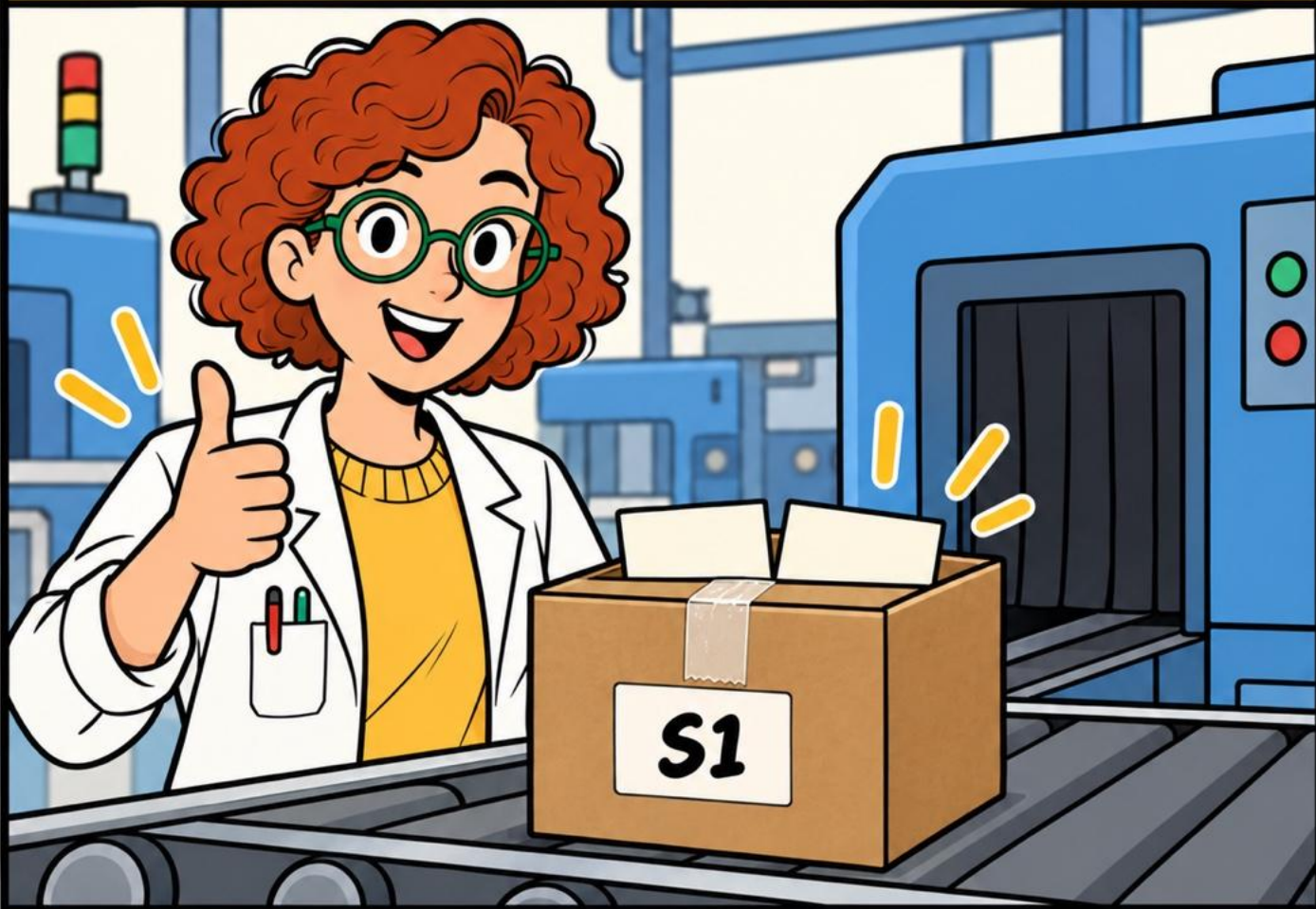
THE PIPELINE FACTORY

Ep 3: The belts



Ep 3

LAST TIME



Each sample is a labelled box: its name plus its two read files, taped together.

Ep 3

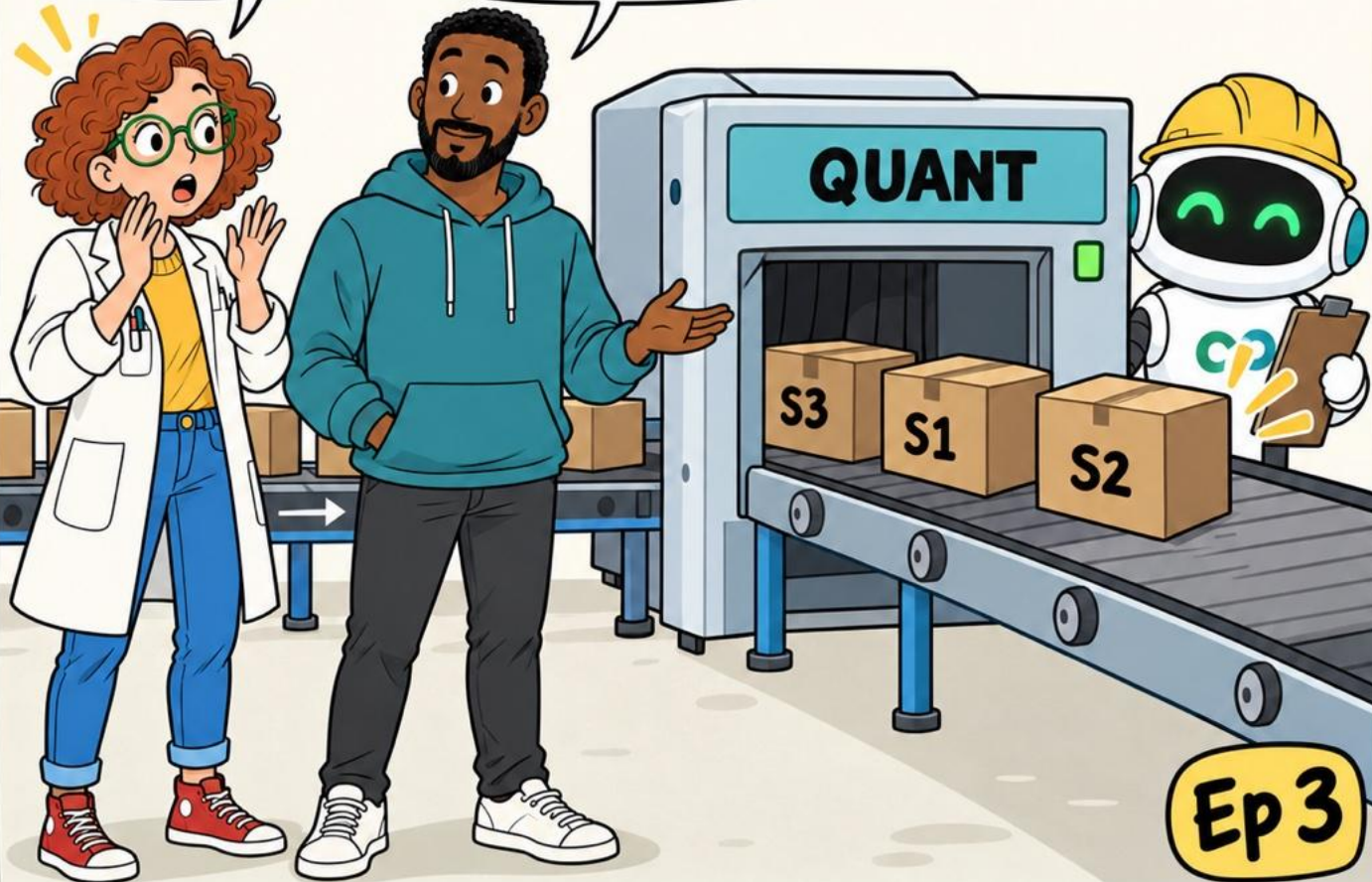
THE NORMAL BELT

(a queue channel)

Why did S3
come out
before S1?

S3 was faster.
Boxes leave in the
order they finish.

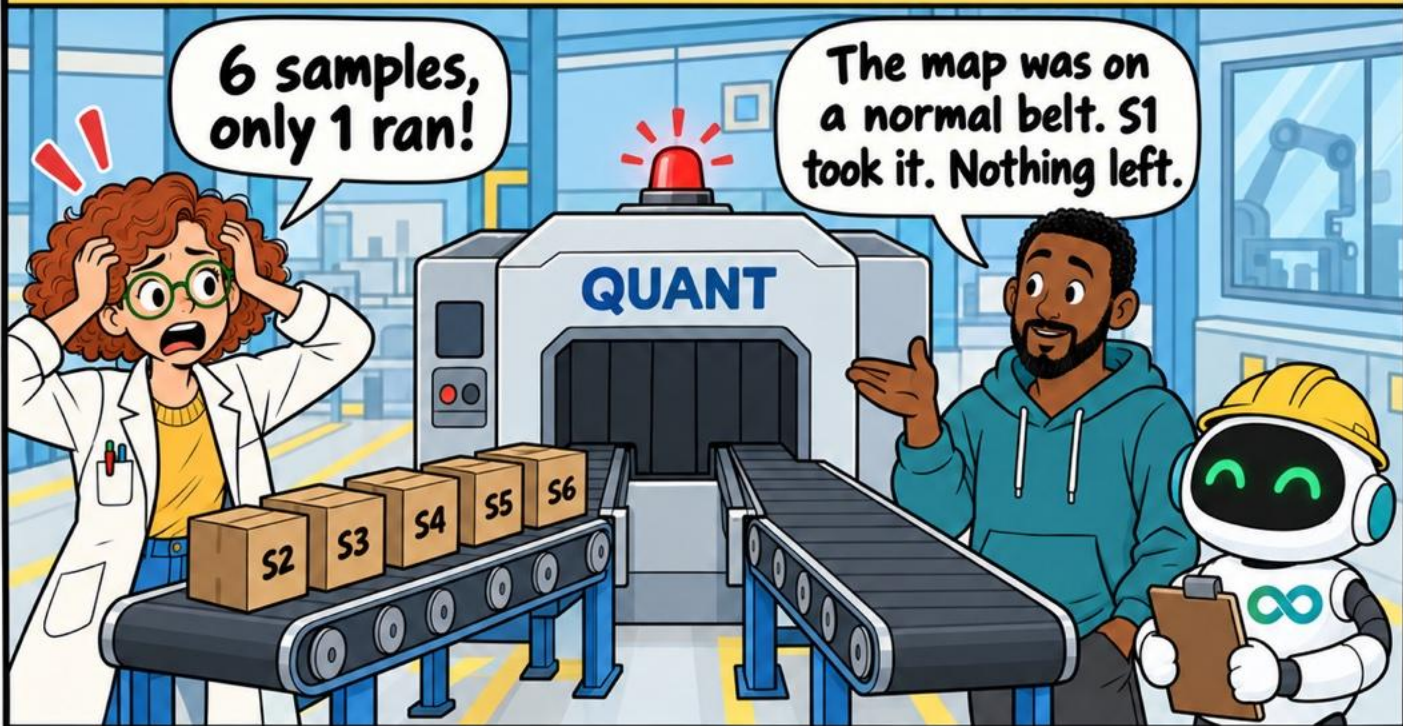
Each box is taken once,
then gone.



ONLY ONE RAN?!

6 samples,
only 1 ran!

The map was on
a normal belt. S1
took it. Nothing left.



?

So a station
stops when a
belt runs
empty?



Exactly.



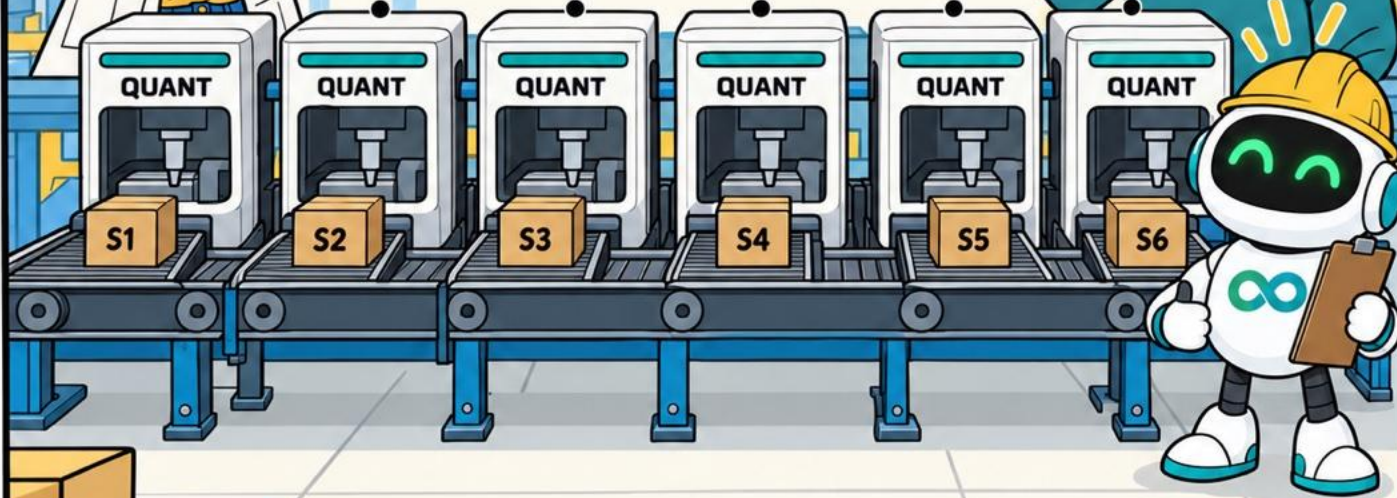
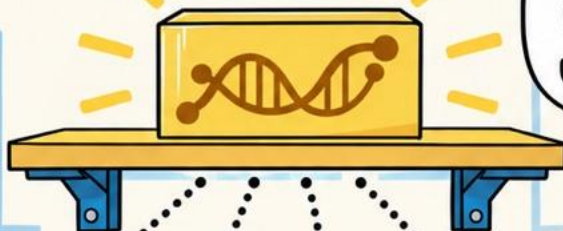
Ep 3

THE SHELF

(a value channel)

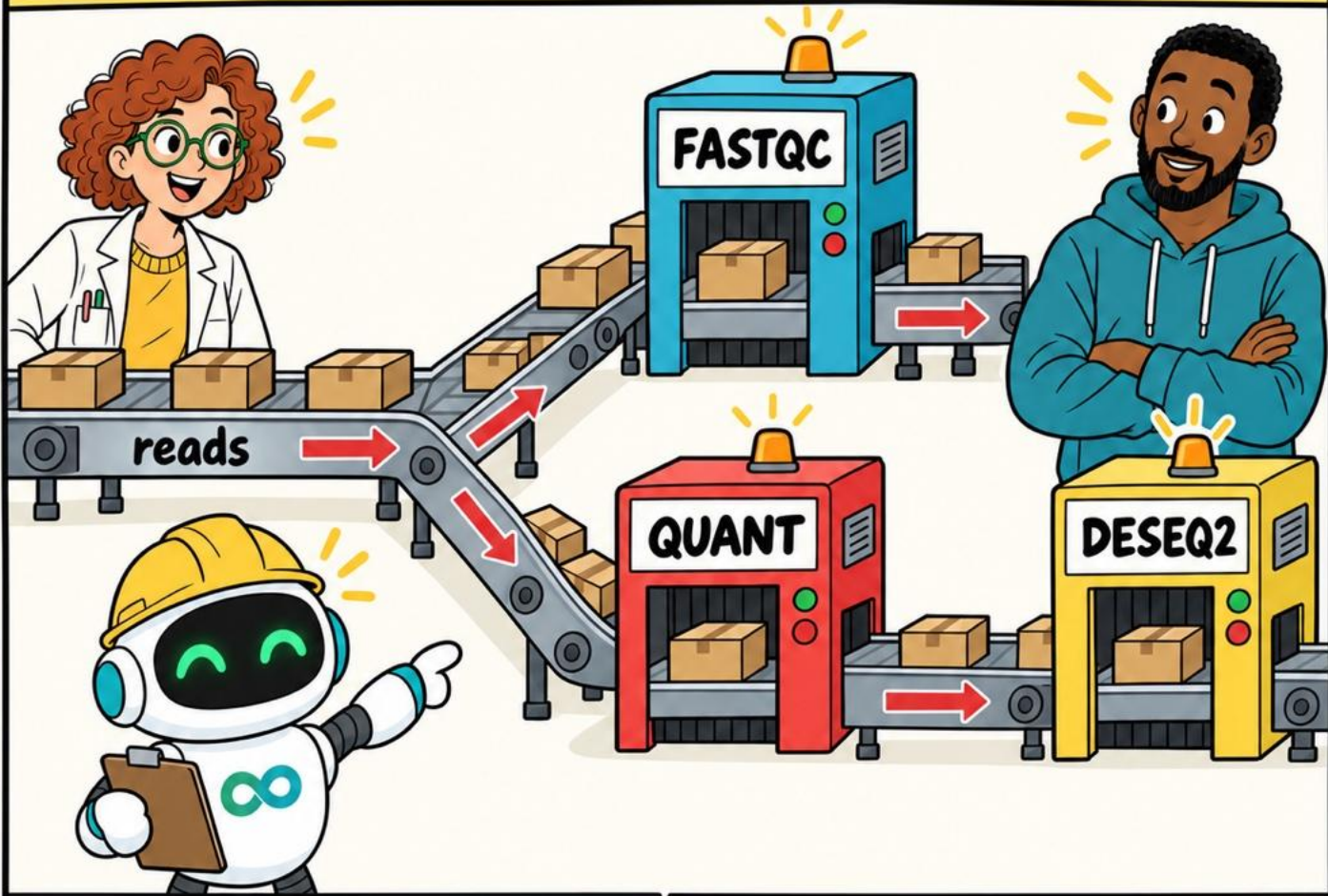
6 samples,
6 runs!

Put the map on a shelf.
Every station run can
use it, again and again.



Ep 3

BELTS CONNECT STATIONS



One belt can feed two stations: reads → FASTQC and QUANT.

A station's output is a new belt: QUANT → DESEQ2.

.collect()

DESEQ2 needs ALL samples to compare groups.

.collect() waits for every box and puts them on one tray.

So DESEQ2 runs once?

Once, on the whole tray.

THE BELTS, IN CODE

```
workflow {  
  // belt: one box per sample (name + 2 read files)  
  reads_ch = channel.fromFilePairs(params.reads, checkIfExists: true)  
  // same belt feeds FASTQC and QUANT  
  FASTQC(reads_ch)  
  // one plain file in -> the map goes on the shelf  
  index_ch = SALMON_INDEX(file(params.transcriptome))  
  // belt: gene counts, one per sample  
  counts_ch = QUANT(reads_ch, index_ch)  
  // .collect(): every box on one tray -> DESEQ2 runs once  
  DESEQ2(counts_ch.collect(), file(params.samplesheet))  
}
```



NEXT TIME

Laptop or cluster?
How much memory?
One room decides.

The control room!

CONTROL ROOM

Ep 4: The control room →

Ep 3



THANKS FOR READING!

Follow me on GitHub,
where I explain how it all works:

<https://github.com/loukesio>